Uncertainty-Aware Multi-Agent Ransomware Detection via Decision MLP Fusion, Score-Space Augmentation, and Bayesian Triage Policy Optimisation

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Abstract—Ransomware detection under operational constraints demands more than a binary verdict: it requires principled uncertainty estimation, heterogeneous evidence fusion, and uncertainty-driven triage that balances autonomous containment against analyst escalation. We present a three-agent framework addressing these requirements through four coupled contributions. First, a Semantic CNN on static PE embeddings, a Behavioural CNN on 16-dimensional hypervisor telemetry, and a Knowledge CNN on ATT&CK tactic embeddings each employ Monte Carlo Dropout to produce per-sample predictive probabilities and epistemic uncertainties. Second, a Decision MLP learns a calibrated aggregate risk score from the six-dimensional score pool, reducing Expected Calibration Error from 0.100 to 0.025 relative to mean fusion—since RanSMAP, RDset, and RansKnow share no real per-sample pairing, the Semantic and Knowledge Agents contribute a fixed, label-independent corpus-mean score during RanSMAP fusion, so calibration rather than incremental AUROC is fusion's principal benefit; Conditional Tabular GAN (CTGAN) augmentation addresses score-space imbalance without retraining base agents. Third, a four-threshold triage policy optimised via Bayesian TPE reduces the escalation rate from 96.8% to 35.5% while raising balanced accuracy from 0.525 to 0.837. Fourth, a Peer Consultation mechanism mitigates knowledge collapse under concurrent benign–ransomware execution, and an Expert Feedback Loop with Elastic Weight Consolidation enables continual adaptation to novel families with negligible catastrophic forgetting (ΔAUPRC=0.001). Evaluated across four generalisation axes over six independent seeds on RanSMAP, RDset, and RansKnow, the framework achieves AUROC 0.996 on static PE detection and AUROC 0.963 on runtime behavioural detection, matched rather than exceeded by the fused Decision MLP output (0.952), which trades a small AUROC margin for substantially better calibration and principled per-sample uncertainty. Under family holdout, AUROC falls to 0.706 (0.676 fused), revealing a meaningful generalisation gap for unseen families; epistemic uncertainty routes these novel-family samples to analyst escalation rather than autonomous containment. A Mondrian conformal prediction wrapper provides distribution-free coverage under exchangeability, auto-classifying 54% of traffic at 99% coverage.

Index Terms—Ransomware detection, multi-agent systems, uncertainty quantification, generative adversarial networks, continual learning, conformal prediction.

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I. Introduction

Ransomware attacks impose escalating operational and financial costs on enterprises and critical infrastructure [1], [2]. State-of-the-art detectors—signature-based antivirus, EDR platforms, or machine-learning classifiers on file-level features—converge on a common architectural limitation: a single scalar confidence score mapped to a binary verdict. When a novel variant or obfuscated execution degrades that score silently, no signal informs the operator that the verdict is unreliable.

We identify three structural gaps in the current state of practice:

1) No principled uncertainty. A softmax output from a deterministic neural network is a point estimate of class membership, not a probability over models. It cannot distinguish a confident correct prediction from a confident incorrect one.
2) Heuristic modality fusion. Combining static PE features with runtime telemetry via equal-weight averaging discards the per-modality reliability signal; a semantic agent operating on encrypted PE sections and a behavioural agent observing obfuscated I/O patterns should not carry equal weight.
3) Binary triage. An alert generated by every uncertain detection overwhelms analysts; equally, routing every uncertain detection to automatic containment risks business disruption.

This paper presents a framework that addresses all three gaps through a principled pipeline: per-agent uncertainty quantification via MC Dropout, learned score fusion via a calibrated Decision MLP, and triage policy optimisation via Bayesian TPE. The framework treats uncertainty as a control variable for deployment rather than a diagnostic afterthought.

Contributions

1) A heterogeneous three-agent architecture in which a Semantic CNN, a Behavioural CNN, and a Knowledge CNN each produce a predictive probability and an epistemic uncertainty via $T = 10$ MC Dropout forward passes.

2) A Decision MLP fusion mechanism with temperature calibration that replaces heuristic mean fusion, reducing ECE from 0.100 to 0.025; because the Semantic and Knowledge Agents contribute a fixed corpus-mean score for RanSMAP samples (no real per-sample pairing exists across the three datasets), this calibration gain—not incremental AUROC over the Behavioural Agent alone—is fusion's principal benefit, supported by CTGAN score-space augmentation to handle class imbalance at the fusion layer.
3) A formal triage policy over $(\hat{p}, \sigma_{\max})$ space with four jointly optimised thresholds via Bayesian TPE, reducing escalation rate from 96.8% to 35.5% while improving balanced accuracy from 0.525 to 0.837.
4) A Peer Consultation mechanism that prevents knowledge collapse under concurrent execution, and an Expert Feedback Loop with EWC enabling continual family adaptation without catastrophic forgetting.

The remainder of this paper reviews related work (Section II), formalises the detection and triage problem (Section III), describes the three agents, MC Dropout uncertainty, Decision MLP fusion, the triage policy, Peer Consultation, and conformal prediction (Sections IV–VIII), presents experimental evaluation (Section IX), and discusses results, limitations, and conclusions (Sections X–XI).

II. Related Work

A. Behavioural Ransomware Detection

Hypervisor-level behavioural monitoring [3], [4] captures low-level I/O and memory access patterns before encryption artefacts surface in the filesystem, providing strong coverage of novel variants, though IRP monitors [5] remain vulnerable to slow-encrypt evasion and API-call classifiers [6], [7] require costly process-level instrumentation. Survey work [8], [9] confirms that runtime behavioural methods outperform static analysis against obfuscated ransomware, but notes a structural limitation shared by all existing systems: they produce a scalar confidence score with no mechanism for expressing per-sample epistemic uncertainty or routing uncertain detections to analyst escalation.

B. Uncertainty Quantification in Detection

Monte Carlo Dropout [10] re-activates dropout at test time, treating each stochastic forward pass as a draw from an approximate posterior over network weights, with the variance of T outputs approximating epistemic uncertainty at $T\times$ the inference cost. Gawlikowski et al. [11] establish MC Dropout and deep ensembles [12] as the most broadly applicable of four neural-network uncertainty method families, though ensembles require retaining M full model copies at inference. For distribution-free guarantees, conformal prediction [13] and its Mondrian variant [14] bound coverage probability without parametric assumptions, generalised to arbitrary risk functions by Angelopoulos and Bates [15]. Uncertainty estimation has been applied to network anomaly detection [16], but ransomware triage—where an autonomous-containment error differs qualitatively in cost from an escalation—has not previously been framed as an uncertainty-conditioned routing problem.

C. Multi-Agent and Ensemble Detection

Ensemble methods for malware classification [17], [18] aggregate multiple classifiers over a shared feature space via majority vote or average-score fusion. Multi-agent intrusion detection systems [19] distribute detection across specialised agents monitoring different network zones, combined through a centralised coordinator. Multi-modal ransomware detection combining static PE features, API-call sequences, and file-system events [8] typically uses equal-weight or attention-based fusion that does not propagate per-modality reliability.

Our framework differs structurally in two respects: the three agents operate on structurally heterogeneous input spaces ($\mathbb{R}^{384}$ PE embeddings, $\mathbb{R}^{16}$ runtime telemetry, $\mathbb{R}^{23}$ ATT&CK vectors) that cannot be concatenated without discarding modality-specific structure, and each agent contributes not only a probability but an epistemic uncertainty, which the Decision MLP uses implicitly to down-weight unreliable modalities and which the orchestrator uses explicitly to route uncertain cases to analyst escalation—a mechanism absent from standard ensemble methods.

D. Generative Augmentation and Continual Learning

Class imbalance in detection datasets is routinely addressed by SMOTE [20] and Conditional Tabular GAN (CTGAN) [21], applied to fraud and intrusion detection [22], [23] and consistently outperforming SMOTE when the minority class is small and non-Gaussian. We apply CTGAN at the score space output of the multi-agent detector: operating on bounded $[0, 1]^6$ probability–uncertainty vectors constrains synthetic samples to the agents' output range, reducing the risk of implausible combinations and eliminating raw-telemetry reconstruction artefacts.

Catastrophic forgetting—gradient updates for a new task systematically overwriting prior weight configurations [24]—is the key barrier to continual adaptation; temporal evaluation frameworks [25]–[27] establish that standard classifiers lose substantial accuracy within months of the training cutoff under concept drift. Elastic Weight Consolidation (EWC) [24] mitigates this via a diagonal Fisher penalty protecting parameters important to retained tasks; our Expert Feedback Loop applies EWC to the Decision MLP, achieving ΔAUPRC=0.001 on retained families after five correction iterations—enabling continual adaptation without retraining base agents.

E. Relationship to Prior Work

This paper extends the two-agent Agentic SABRE framework [28] (Semantic + Behavioural CNN, MC

Dropout, fixed triage policy) with four technically distinct contributions: (1) a third Knowledge CNN agent on ATT&CK tactic embeddings, expanding the score pool from $[0,1]^4$ to $[0,1]^6$; (2) Bayesian TPE optimisation of four jointly tuned triage thresholds $(\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ replacing the fixed (τ=0.10, κ=0.10) pair; (3) CTGAN score-space augmentation at the fusion layer; and (4) EWC continual learning and Mondrian conformal prediction wrappers. All experiments and reported numerical results in this paper were rerun independently; no numerical results or figures are reused from [28], though both papers draw on the same underlying RDset and RanSMAP datasets.

III. Problem Formulation

A. Detection Task

Let $\mathcal{X} = \mathcal{X}_{\text{sem}} \times \mathcal{X}_{\text{beh}} \times \mathcal{X}_{\text{know}}$ denote the joint input space of three heterogeneous modalities, and let $\mathcal{Y} = \{0, 1\}$ denote the binary ransomware label. A sample is observed as a triple $\mathbf{x} = (\mathbf{x}_{\text{sem}}, \mathbf{x}_{\text{beh}}, \mathbf{x}_{\text{know}}) \in \mathcal{X}$, where

- $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ is a sentence-transformer embedding of Portable Executable (PE) section names;
- $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ is a 16-dimensional feature vector extracted from a $T_{\text{window}} = 10\,\text{s}$ telemetry window (see Section IV);
- $\mathbf{x}_{\text{know}} \in \mathbb{R}^{d}$ is an embedding of ATT&CK tactic profiles derived from threat-intelligence transcripts.

B. Triage Task

Rather than mapping $\mathbf{x}$ to a binary label, the triage task maps $\mathbf{x}$ to one of three actions $\mathcal{A}$ = {Auto-Contain, Escalate, Allow} according to a policy $\pi : (\hat{p}, \sigma_{\max}) \to \mathcal{A}$, where $\hat{p} \in [0,1]$ is the fused risk score and $\sigma_{\max} \in [0,1]$ is the maximum per-agent epistemic uncertainty.

Definition 1 (Triage Policy). Given thresholds $\Theta = (\tau_{\text{high}}, \tau, \kappa, \kappa_{\text{low}})$ with $0 < \tau \le \tau_{\text{high}} \le 1$ and $0 < \kappa_{\text{low}} \le \kappa \le 1$, the triage policy is:

$$\pi(\hat{p}, \sigma_{\max}; \Theta) = \begin{cases} \text{Auto-Contain} & \hat{p} \ge \tau_{\text{high}} \ \wedge \ \sigma_{\max} \le \kappa_{\text{low}}, \\ \text{Escalate} & \hat{p} \ge \tau \ \vee \ \sigma_{\max} > \kappa, \\ \text{Allow} & \text{otherwise.} \end{cases} \tag{1}$$

Since $\tau \le \tau_{\text{high}}$, the Auto-Contain and Escalate conditions overlap on the region $\hat{p} \ge \tau_{\text{high}} \wedge \sigma_{\max} \le \kappa_{\text{low}}$: this is not an ambiguity but a stated precedence order, evaluated top-to-bottom, so Auto-Contain takes priority whenever both conditions hold.

C. Optimisation Objective

Let bal_acc(Θ) denote the balanced accuracy of the joint detection–triage system under thresholds Θ, and let esc_rate(Θ) denote the fraction of samples routed to Escalate. The policy optimisation problem is:

$$\Theta^* = \arg\max_{\Theta \in \mathcal{F}} \ J(\Theta) := \text{bal_acc}(\Theta) - \lambda\, \text{esc_rate}(\Theta), \tag{2}$$

where $\lambda \ge 0$ is an operational cost parameter encoding the relative burden of analyst escalation, and $\mathcal{F}$ is the feasibility region $\{\Theta : \tau \le \tau_{\text{high}},\ \kappa_{\text{low}} \le \kappa\}$.

Remark 1. The parameter λ is a deployment decision, not a modelling one. An organisation with constrained analyst capacity sets higher λ; one prioritising missed-detection avoidance sets $\lambda \approx 0$.

IV. Agent Architecture and Uncertainty Quantification

A. Monte Carlo Dropout as Approximate Bayesian Inference

Let $f_\theta : \mathcal{X}_i \to [0,1]$ denote a neural network with parameters θ trained on modality i; standard dropout is disabled at inference, yielding a deterministic point estimate. MC Dropout [10] instead keeps dropout active at test time, treating each stochastic forward pass as a draw from an approximate posterior $q_\phi(\theta) \approx p(\theta \mid \mathcal{D})$.

Formally, for T stochastic forward passes with dropout masks $\{\mathbf{z}_t\}_{t=1}^{T}$:

$$\hat{p}_i(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} f_{\theta \odot \mathbf{z}_t}(\mathbf{x}), \tag{3}$$

$$\sigma_i(\mathbf{x}) = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T} \left(f_{\theta \odot \mathbf{z}_t}(\mathbf{x}) - \hat{p}_i(\mathbf{x}) \right)^2}, \tag{4}$$

where $\odot$ denotes element-wise masking. The quantity σ_i captures epistemic uncertainty—reducible uncertainty arising from limited training data—as distinct from aleatoric uncertainty inherent to the data generating process [29].

Proposition 1 (Epistemic Signal). When a test sample $\mathbf{x}$ lies far from the training manifold of agent i (i.e., a novel ransomware family), we hypothesise that the variance $\sigma_i^2(\mathbf{x})$ tends to increase with distributional shift $d(\mathbf{x}, \mathcal{D}_{\text{train}})$ [10]. Approximate Bayesian methods do not provide a formal monotonicity guarantee in general; we verify this tendency empirically across family and hardware holdout splits (Supplementary Section S-VI).

We use $T = 10$ passes per agent, empirically verified to stabilise σ_i to within 1% of the $T = 50$ estimate (Section IX-O); Supplementary Section S-VI quantifies this stability and the associated latency trade-off.

B. Semantic Agent

The Semantic Agent operates on static PE structure. Given a PE file, section names are concatenated and encoded via a frozen sentence transformer (all-MiniLM-L6-v2) to produce $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$. A Semantic CNN f_{sem} with MC Dropout maps this embedding to $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$.

Training data: RDset [30], [31], comprising 138,047 PE files (41,123 ransomware; 96,924 benign) with SMOTE balancing applied at the feature level prior to training.

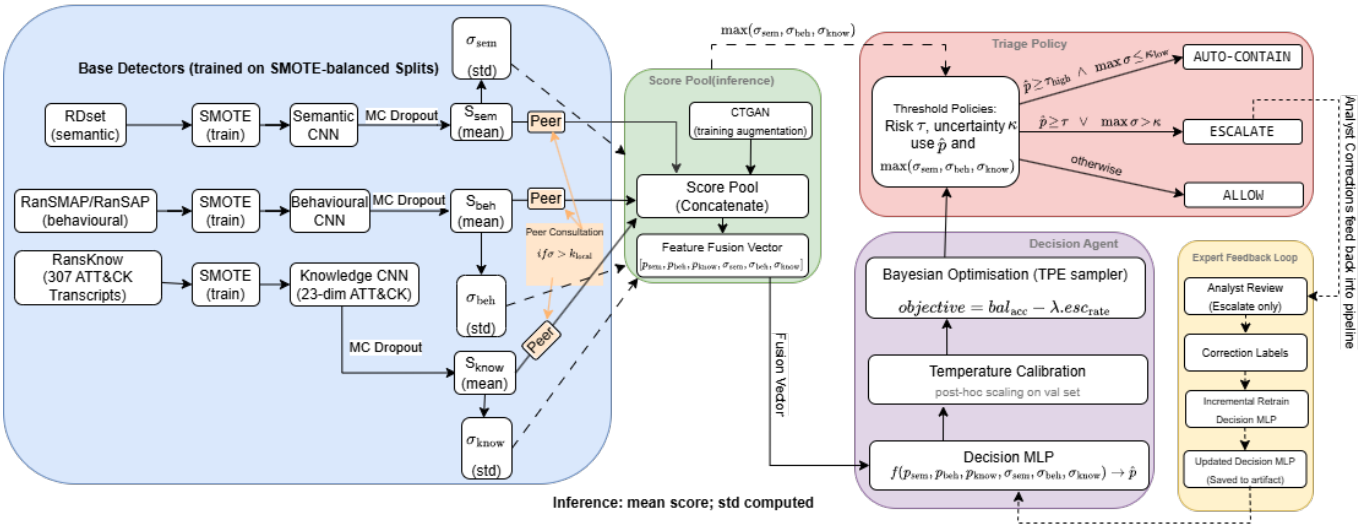


Fig. 1. End-to-end framework pipeline. Three specialised CNN agents (Semantic, Behavioural, Knowledge) each run T=10 MC Dropout forward passes to produce a probability–uncertainty pair $(\hat{p}_i, \sigma_i)$. Peer Consultation adjusts any agent whose σ_i exceeds its local threshold before the six-dimensional score pool is assembled. The Decision MLP fuses the pool into a calibrated risk score; Bayesian TPE derives the four-threshold triage policy that maps $(\hat{p}, \sigma_{\max})$ to Auto-Contain, Escalate, or Allow. Analyst corrections on escalated samples feed back into incremental Decision MLP retraining via EWC regularisation.

C. Behavioural Agent

The Behavioural Agent operates on runtime storage and memory access patterns collected via a thin BitVisor hypervisor [4]. Within each $T_{\text{window}} = 10\,\text{s}$ window we compute a $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ feature vector partitioned as $\mathbf{x}_{\text{beh}} = (\chi_s, \chi_m)$, where:

Storage features $\chi_s \in \mathbb{R}^5$:

$$\chi_s = \Big(\bar{T}_{s,r},\ \bar{T}_{s,w},\ V_{s,r},\ V_{s,w},\ \bar{H}_{s,w}\Big), \tag{5}$$

where $\bar{T}_{s,r}$ and $\bar{T}_{s,w}$ are mean read/write throughputs [byte/s]; $V_{s,r}$ and $V_{s,w}$ are variances of accessed Logical Block Addresses (LBA); and $\bar{H}_{s,w}$ is mean Shannon entropy of written sectors, computed as $H(X) = -\sum_i p(x_i)\log_2 p(x_i)$.

Memory features $\chi_m \in \mathbb{R}^{11}$:

$$\chi_m = \Big(H_{m,w},\ H_{m,rw},\ C_{4K,r},\ C_{4K,w},\ C_{4K,rw},\ C_{4K,x},\ C_{2M,r},\ C_{2M,rw},\ C_{\text{MMIO},r},\ C_{\text{MMIO},w},\ V_{m,w}\Big), \tag{6}$$

where $H_{m,w}$ and $H_{m,rw}$ are Shannon entropies of written and read/written memory pages; $C_{s,o}$ denotes the count of memory accesses of size $s \in \{4\,\text{KiB}, 2\,\text{MiB}, \text{MMIO}\}$ and operation $o \in \{r, w, rw, x\}$; and $V_{m,w}$ is the variance of written Guest Physical Addresses (GPA).

Training data: RanSMAP [4], comprising 35,857 ten-second windows from 1,776 sessions across 15 hardware configurations, six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip).

D. Knowledge Agent

The Knowledge Agent encodes ATT&CK tactic profiles derived from threat-intelligence transcripts (RansKnow [32]: 307 ATT&CK-aligned transcripts). A Discourse CNN f_{know} maps the tactic embedding $\mathbf{x}_{\text{know}} \in \mathbb{R}^d$ to $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$.

E. Evaluation Protocol and Generalisation Splits

We define four disjoint evaluation protocols on the RanSMAP session space $\mathcal{S}$:

1) Random: 80/20 stratified split over $\mathcal{S}$.
2) Family holdout (more precisely a class holdout: the held-out set spans both ransomware families and benign applications): $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{family}(s) \in \{\text{REvil}, \text{WannaCry}, \text{Firefox}, \text{Zip}\}\}$. We retain the shorter name "family holdout" throughout for brevity.
3) Hardware holdout: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : \text{hw}(s) \in \{\text{i5-gen12}, \text{i7-gen12}, \text{i3-gen12/DDR5}\}\}$.
4) Temporal: $\mathcal{S}_{\text{test}} = \{s \in \mathcal{S} : t(s) \geq t_{0.85}\}$, where $t_{0.85}$ is the 85th percentile timestamp.

All results are reported as mean ± std over $K = 6$ independent random seeds.

V. Decision MLP Fusion and Score-Space Augmentation

A. Score Pool Construction

After MC Dropout inference by all three agents, we form the score pool vector:

$$\mathbf{v}(\mathbf{x}) = \big(\hat{p}_{\text{sem}},\ \sigma_{\text{sem}},\ \hat{p}_{\text{beh}},\ \sigma_{\text{beh}},\ \hat{p}_{\text{know}},\ \sigma_{\text{know}}\big) \in [0,1]^6. \tag{7}$$

The maximum per-agent uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$ is derived from $\mathbf{v}$ and used independently in the triage policy (Section VI), but is not fed to the Decision MLP.

TABLE I
Which dataset each agent is trained and evaluated on, and what it contributes to a RanSMAP fusion score $\mathbf{v}(\mathbf{x})$ (Remark 2). Only the Behavioural Agent's contribution to $\mathbf{v}$ varies per RanSMAP sample.

Agent	Trained & evaluated on	Contribution to RanSMAP $\mathbf{v}(\mathbf{x})$
Semantic	RDset (own train/test split)	Fixed corpus-mean $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$, identical $\forall\, \mathbf{x}$
Behavioural	RanSMAP (own train/test split)	Per-sample $(\hat{p}_{\text{beh}}(\mathbf{x}), \sigma_{\text{beh}}(\mathbf{x}))$
Knowledge	RansKnow (own train/test split)	Fixed corpus-mean $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$, identical $\forall\, \mathbf{x}$

Remark 2 (Cross-Dataset Fusion Without Per-Sample Pairing). RDset, RanSMAP, and RansKnow are independently collected: no executable appears in more than one dataset, so a RanSMAP execution window has no corresponding PE binary or ATT&CK transcript. We therefore do not attempt to condition $\hat{p}_{\text{sem}}$ or $\hat{p}_{\text{know}}$ on the RanSMAP sample's own (family) identity—doing so via a per-family lookup keyed by the sample's true label would leak that label into the score pool. Instead, for every RanSMAP evaluation, $(\hat{p}_{\text{sem}}, \sigma_{\text{sem}})$ and $(\hat{p}_{\text{know}}, \sigma_{\text{know}})$ are fixed, label-independent constants: the MC Dropout mean and standard deviation of each agent's own corpus (RDset for Semantic, RansKnow for Knowledge), computed once and reused for every RanSMAP sample regardless of family. Consequently, in the RanSMAP evaluations of Section IX, only the Behavioural Agent carries per-sample discriminative signal; the Semantic and Knowledge Agents contribute a fixed prior that the Decision MLP can learn to weight, and Peer Consultation (Section VII) can still substitute when σ_{know} or σ_{sem} exceeds a peer's, but neither agent can improve fused AUROC over the Behavioural Agent alone in this setting. This is why the Decision MLP's contribution below is measured in calibration and uncertainty-gated triage quality rather than in raw discrimination. Table I makes this mapping explicit.

Because the Semantic and Knowledge Agents' own in-domain AUROC (RDset 0.996; Knowledge Agent metrics not separately reported given RansKnow's small size, Section X) says nothing about their contribution to a RanSMAP fusion score—which is fixed and sample-independent by construction—the framework in its RanSMAP evaluations (Section IX) is honestly a Behavioural–Knowledge–Semantic score-pool fusion with only the Behavioural Agent contributing per-sample signal, not a validated three-way sample-level fusion. The Semantic Agent's own three-way-independent contribution is validated separately, in-domain, on RDset.

Remark 3 (Why Score Space). Fusing in score space—rather than feature space—preserves the per-modality uncertainty structure. Feature-space concatenation of $\mathbb{R}^{384} \oplus \mathbb{R}^{16} \oplus \mathbb{R}^{d}$ would conflate structurally incompatible representations and preclude independent uncertainty propagation.

B. CTGAN Score-Space Augmentation

Let $\mathcal{V}_{\text{train}} = \{(\mathbf{v}_j, y_j)\}_{j=1}^{N}$ denote the score-pool training set with labels $y_j \in \{0, 1\}$, where the ransomware class is a fraction $r \ll 0.5$ of samples. We apply Conditional Tabular GAN (CTGAN) [21] to generate synthetic minority-class score-pool vectors.

The CTGAN minimax objective is:

$$\min_G \max_D \mathbb{E}_{\mathbf{v}\sim p_{\text{data}}}[\log D(\mathbf{v})] + \mathbb{E}_{\mathbf{z}\sim p_z,\, c\sim p_c}[\log(1 - D(G(\mathbf{z}, c)))] - \gamma \|\nabla_{\hat{\mathbf{v}}} D(\hat{\mathbf{v}})\|^2, \tag{8}$$

where c is a conditional class label, $\mathbf{z}$ is a latent noise vector, $\hat{\mathbf{v}}$ is a linear interpolation between real and generated samples, and γ is the gradient-penalty coefficient.

Crucially, CTGAN operates on $\mathbf{v} \in [0, 1]^6$, not on raw features: this ensures that the generator cannot produce physically impossible score combinations and preserves the calibration structure of the base agents.

C. Decision MLP Architecture

The Decision MLP $g_\phi : [0, 1]^6 \to [0, 1]$ maps the score pool to a fused risk score:

$$g_\phi(\mathbf{v}) = \sigma(\mathbf{W}_2\, \text{ReLU}(\mathbf{W}_1 \mathbf{v} + \mathbf{b}_1) + \mathbf{b}_2)\,, \tag{9}$$

with dimensions $6 \to 32 \to 1$ and sigmoid output. Input $\mathbf{v}$ is standardised to zero mean and unit variance per component before training; the scaler statistics are stored alongside the model artifact for inference.

D. Temperature Calibration

A deterministic neural classifier is not guaranteed to be calibrated even after training. We apply temperature scaling [33]: the logit $\ell = \log(\hat{p}/(1 - \hat{p}))$ is divided by a learned scalar $T > 0$ before the final sigmoid:

$$\hat{p}_{\text{cal}} = \sigma(\ell / T), \quad T^* = \arg\min_T \text{NLL}_T(\mathcal{D}_{\text{cal}}). \tag{10}$$

Expected Calibration Error is defined as:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|, \tag{11}$$

where $\{B_m\}_{m=1}^{M}$ partition $[0, 1]$ into M equal-width confidence bins, $\text{acc}(B_m)$ is empirical accuracy and $\text{conf}(B_m)$ is mean predicted confidence within bin B_m.

After calibration, ECE decreases from 0.100 (mean fusion baseline) to 0.025 (Decision MLP with temperature scaling).

VI. Triage Policy and Bayesian Optimisation

A. Triage Policy in $(\hat{p}, \sigma_{\max})$ Space

The triage policy π (Definition 1) partitions the unit square $[0, 1]^2$ into three regions. The fused risk $\hat{p} = g_\phi(\mathbf{v})$ and maximum agent uncertainty $\sigma_{\max}$ jointly determine the action. The boundary conditions ensure that:

- Auto-Contain requires both high risk and low uncertainty—autonomous action is taken only when the system is jointly confident in detection and reliable;

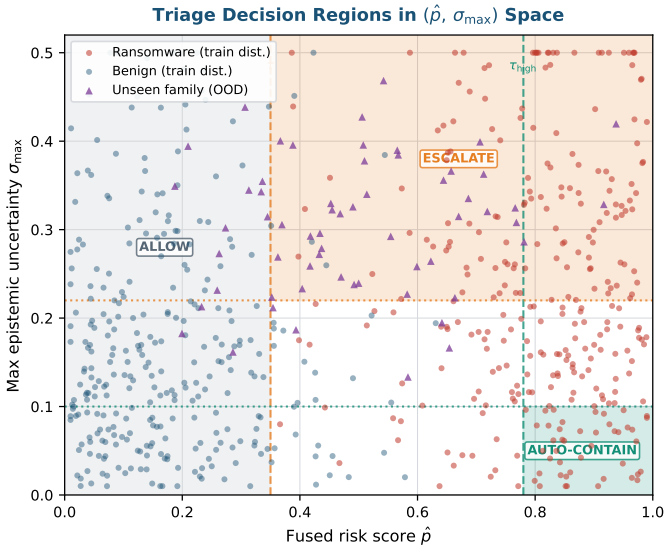


Fig. 2. Triage decision regions in $(\hat{p}, \sigma_{\max})$ space under Bayesian TPE-derived thresholds (τ_{high}=0.78, τ=0.35, κ=0.22, κ_{low}=0.10). Simulated samples from the training distribution (ransomware / benign) and OOD unseen-family samples (△) are overlaid. OOD samples cluster in the Escalate region due to elevated $\sigma_{\max}$.

- Escalate fires when either the risk score is elevated ($\hat{p} \geq \tau$) or agent disagreement is high ($\sigma_{\max} > \kappa$), routing borderline and uncertain samples to an analyst rather than acting autonomously;
- Allow covers the confident-benign region.

Fig. 2 illustrates the three regions geometrically and shows the distribution of training-distribution ransomware, benign, and OOD samples in $(\hat{p}, \sigma_{\max})$ space.

B. Bayesian Triage Policy Optimisation via TPE

The objective $J(\Theta)$ in (2) is non-differentiable in Θ (discrete triage counts), so we employ Bayesian optimisation via Tree-structured Parzen Estimation (TPE) [34].

TPE models the probability density of J conditioned on Θ as:

$$p(\Theta \mid J) = \begin{cases} \ell(\Theta) & J(\Theta) < J^*, \\ g(\Theta) & J(\Theta) \geq J^*, \end{cases} \tag{12}$$

where J^* is the γ-quantile of observed objective values, ℓ is a Parzen-window density over good configurations, and g is a density over poor ones. The acquisition criterion selects:

$$\Theta_{\text{next}} = \arg\max_{\Theta} \frac{\ell(\Theta)}{g(\Theta)}, \tag{13}$$

which approximates the Expected Improvement criterion.

We run 300 TPE trials over $\mathcal{F}$ with $\lambda = 0.3$; the search budget is chosen based on the dimensionality of Θ and the cost of evaluating J on the validation set (approximately 0.15 s per trial). Results and convergence analysis are presented in Section IX-G.

TABLE II
Triage policy comparison on random-stratified test split (mean ± 95% CI over K=6 seeds). Bal. acc = balanced accuracy; Esc = escalation rate; AC = auto-contain rate. ↑ = higher is better; ↓ = lower is better.

Policy	Bal. acc ↑	Esc (%) ↓	AC (%) ↑
SABRE+ Fixed [28]	0.525 ± 0.001	96.8 ± 0.1	0.6 ± 0.0
Grid search	0.843 ± 0.001	58.7 ± 0.2	2.6 ± 0.0
Bayesian TPE (300 trials)	$\mathbf{0.837 \pm 0.001}$	$\mathbf{35.5 \pm 0.5}$	$\mathbf{26.7 \pm 0.4}$

C. Comparison with Grid Search and Fixed Thresholds

Table II summarises the triage outcome under the three policies on the random-stratified test split. Bayesian TPE cuts the escalation rate relative to grid search (35.5% vs 58.7%) at slightly lower balanced accuracy (0.837 vs 0.843), and increases the auto-contain rate more than tenfold (26.7% vs 2.6%). Fixed thresholds (τ=0.10, κ=0.10, inherited from [28]) are wholly unsuitable for autonomous deployment: 96.8% escalation saturates analyst capacity, and 0.525 balanced accuracy reflects near-random triage.

VII. Peer Consultation and Expert Feedback Loop

A. Peer Consultation

Under concurrent benign–ransomware execution, the Behavioural Agent observes superimposed access patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$. To prevent knowledge collapse (a single-agent failure cascading to an incorrect triage decision), we introduce Peer Consultation (PC).

Definition 2 (Peer Consultation). When agent i's epistemic uncertainty $\sigma_i(\mathbf{x}) > \kappa_{\text{PC}}$ for a threshold $\kappa_{\text{PC}} > 0$, agent i's probability only is replaced by a weighted combination of its peer agents' current probabilities:

$$\hat{p}_i^{\text{PC}}(\mathbf{x}) = \sum_{j \neq i} w_j \, \hat{p}_j(\mathbf{x}), \tag{14}$$

where weights are proportional to the inverse uncertainty of each peer, $w_j \propto 1/\sigma_j$, normalised so that $\sum_{j \neq i} w_j = 1$. Agent i's own uncertainty σ_i is not modified by this substitution—it is used unchanged by the triage policy (Section VI), so a consulted agent still contributes its original (elevated) uncertainty to $\sigma_{\max}$. Each agent maintains its own threshold $\kappa_{\text{PC}}^{(i)}$; in practice all are set equal to the triage threshold κ optimised by Bayesian TPE.

The three triggering conditions are evaluated independently, in the fixed order Semantic → Behavioural → Knowledge, and each substitution uses whatever probabilities are current at that point in the sequence—so if two or three agents are simultaneously uncertain, a later agent's replacement can draw on an earlier agent's already-substituted probability rather than its original one. For example, if both σ_{sem} and σ_{beh} exceed κ_{PC}, the Semantic Agent's probability is substituted first (from the Behavioural and Knowledge Agents' original probabilities), and the Behavioural Agent's substitution then draws on the Semantic Agent's already-substituted probability together with the Knowledge Agent's original one. Peer

Consultation does not suppress the triggering agent's uncertainty; rather, it routes the sample to the Escalate action while ensuring the score pool reflects available evidence from the more reliable modalities.

B. Expert Feedback Loop with EWC

The Expert Feedback Loop allows an analyst-approved correction label $\tilde{y}$ for an escalated sample to update the Decision MLP g_ϕ incrementally via Elastic Weight Consolidation (EWC) [24], without catastrophic forgetting of the base agents' knowledge. Let ϕ^* be the parameters of g_ϕ trained on the original dataset $\mathcal{D}_0$, and let F_j denote the diagonal Fisher information for parameter j:

$$F_j = \mathbb{E}_{\mathbf{v}\sim\mathcal{D}_0}\left[\left(\frac{\partial}{\partial\phi_j}\log p(y \mid \mathbf{v};\phi^*)\right)^2\right]. \tag{15}$$

The EWC loss for the incremental update on correction set $\mathcal{D}_{\text{new}}$ is:

$$\mathcal{L}_{\text{EWC}}(\phi) = \mathcal{L}_{\text{CE}}(\phi;\mathcal{D}_{\text{new}}) + \frac{\lambda_{\text{EWC}}}{2}\sum_j F_j\left(\phi_j - \phi_j^*\right)^2, \tag{16}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss on new samples and λ_{EWC} controls the trade-off between plasticity and stability. We measure catastrophic forgetting as:

$$\Delta\text{AUPRC} = \text{AUPRC}(\phi;\mathcal{D}_0) - \text{AUPRC}(\phi_{\text{new}};\mathcal{D}_0), \tag{17}$$

achieving $\Delta\text{AUPRC} = 0.001$ on retained families after five feedback iterations.

VIII. Conformal Prediction Wrapper

A. Mondrian Conformal Prediction

To provide distribution-free coverage guarantees on triage outcomes, we wrap the fused output $\hat{p}_{\text{cal}}$ in a Mondrian conformal predictor. Let $\alpha \in (0,1)$ be the target miscoverage rate.

Definition 3 (Conformal Prediction Set). Given a calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{v}_i, y_i)\}_{i=1}^n$, define non-conformity scores $s_i = 1 - f_y(\mathbf{v}_i)$ for the true class y_i. The conformal prediction set at level α for a new sample $\mathbf{v}$ is:

$$\mathcal{C}_\alpha(\mathbf{v}) = \left\{y \in \mathcal{Y} : s(\mathbf{v},y) \le Q_{1-\alpha}(\{s_i\}_{i=1}^n)\right\}, \tag{18}$$

where $Q_{1-\alpha}$ is the $(1-\alpha)$-quantile of calibration scores.

Theorem 1 (Marginal Coverage Guarantee). Under exchangeability of calibration and test samples:

$$\mathbb{P}(Y \in \mathcal{C}_\alpha(\mathbf{v})) \ge 1 - \alpha. \tag{19}$$

Mondrian conformal prediction stratifies the calibration non-conformity scores by each calibration sample's true label (standard label-conditional Mondrian CP), conditioning (18) separately for each class. Theorem 1 provides a formal marginal coverage guarantee under exchangeability of calibration and test samples; Mondrian stratification yields class-conditional coverage empirically but does not carry a separate distribution-free guarantee beyond the marginal bound. The exchangeability assumption holds for the random-stratified split; under temporal and family holdout, the formal guarantee does not apply and coverage is verified empirically (Section IX-L).

IX. Experimental Evaluation

A. Datasets

RDset [30], [31]: 138,047 PE files (41,123 ransomware; 96,924 benign; ≈30% prevalence), used exclusively for Semantic Agent training/evaluation with disjoint train/test splits at the binary (file-hash) level.

RanSMAP [4]: 35,857 ten-second execution windows from 1,776 sessions across 15 hardware configurations (four CPUs: Celeron G3920, Core i3-12100, Core i5-12500, Core i7-12700; six DDR4/DDR5 frequencies; two RAM capacities), six ransomware families (Conti, Darkside, LockBit, REvil, Ryuk, WannaCry), and six benign applications (AESCrypt, Firefox, Idle, Office, SDelete, Zip) under BitVisor hypervisor surveillance; moderate class imbalance addressed via SMOTE at the feature level. A separate variants partition provides 1,397 windows from seven Conti sub-variants for OOD analysis; a mix partition provides 600 windows from ten concurrent execution combinations. Neither is used for training.

RansKnow [32]: 307 ATT&CK-aligned threat-intelligence transcripts, used exclusively for Knowledge Agent training. Score-space imbalance at the fusion layer is addressed separately by CTGAN augmentation (Section V).

B. Implementation Details

Agent architectures. All three base agents share one lightweight single-block design, scaled to their respective input dimensionality: a single-channel input sequence (of length 384, 16, or 23 respectively—the input dimensionality, not a channel count) through one Conv1d(1→C, k=3, same-padding)–ReLU–MaxPool1d(2)–Dropout(p=0.3) block, Flatten, FC(→ 64 or 128)–ReLU, FC(→ 1)–Sigmoid. The Semantic CNN takes $\mathbf{x}_{\text{sem}} \in \mathbb{R}^{384}$ with C=32 channels and a 128-unit hidden FC layer; SMOTE is applied to RDset at the embedding level (r_{minority}=0.30). The Behavioural CNN takes $\mathbf{x}_{\text{beh}} \in \mathbb{R}^{16}$ with C=16 channels and a 64-unit hidden FC layer (8,385 parameters, 32.8 KB); SMOTE is applied to RanSMAP training windows. The Knowledge CNN mirrors this architecture scaled to $\mathbf{x}_{\text{know}} \in \mathbb{R}^{23}$, trained on 307 RansKnow transcripts (70/15/15 train/val/test by family label). None of the three base agents use BatchNorm. The Decision MLP maps $\mathbf{v} \in [0,1]^6$ (standardised per component) through FC(6→32)–ReLU–Dropout(p=0.2), FC(32→1), Sigmoid, with temperature scaling applied post-hoc via L-BFGS on the validation set. Full architecture pseudocode is given in Supplementary Algorithms S1–S2.

Training and hyperparameters. All networks use Adam (η=10^{-3}, batch size 64, up to 50 epochs, early stopping patience 6, best weights restored), K=6 seeds for all RanSMAP experiments. CTGAN, Bayesian TPE, and

EWC hyperparameters are given with their respective results in Sections IX-G and IX-M and Supplementary Algorithm S2 (ctgan v0.9, Optuna 3.5). Hardware: NVIDIA GB10 GPU, Python 3.10, PyTorch 2.2; inference latency measured over 500 repetitions (50 warm-up, batch size 32).

Out-of-fold Decision MLP training features. Scoring the Behavioural Agent's own training windows with a checkpoint trained on those same windows would let the Decision MLP fit an optimistically confident score it never sees again at test time. We therefore build the Behavioural component of every Decision MLP training vector out-of-fold: the random-stratified training sessions are partitioned into five session-grouped, stratified folds (StratifiedGroupKFold), a Behavioural CNN is trained on each set of four folds, and the held-out fifth fold is scored with that fold's own checkpoint; concatenating all five folds' held-out scores covers the full training set with no sample scored by a model that trained on it. The leave-one-family-out evaluation (Section IX-S) applies the same five-fold cross-fitting within each family-excluded training pool, so its Decision MLP training features are simultaneously family-excluded and out-of-fold; test-time scoring for each held-out family uses a Behavioural checkpoint trained on that family-excluded pool, never on the held-out family. Test-time inference throughout uses the full-training-set checkpoint on genuinely held-out test sessions and requires no cross-fitting.

C. Baseline Comparisons

We compare against:

1) Hirano and Kobayashi [4]: 1D-CNN + LSTM baseline on RanSMAP (23D features, $T_{\text{window}} = 0.1\,\text{s}$, $T_d = 30\,\text{s}$) — direct dataset comparison.
2) Mean fusion: [B+K] equal-weight average of $\hat{p}_{\text{beh}}, \hat{p}_{\text{know}}$ (Semantic excluded), without the Decision MLP.
3) Fixed hand-tuned triage thresholds Θ_{fixed}.
4) Grid search over Θ.
5) Fusion-architecture alternatives to the one-layer Decision MLP: a two-layer MLP, Random Forest, and Gradient Boosting, all trained on the identical score pool (Section IX-Q).

D. Main Results: Agent-Level AUROC

Table III reports AUROC for each agent modality and the Decision MLP fusion across four evaluation splits. The Semantic Agent (0.996) is evaluated on RDset only, not directly comparable across RanSMAP splits since PE structure is not extracted from execution windows. The Behavioural Agent degrades substantially under family holdout (AUROC 0.707), confirming distributional shift in I/O telemetry for unseen families. Because the Semantic and Knowledge Agents contribute only a fixed corpus-mean score during RanSMAP fusion (Remark 2), the Decision MLP cannot exceed the Behavioural Agent's own discrimination on any split (random −0.011; hardware −0.011; temporal −0.009; family −0.030); mean fusion degrades further still on every split (e.g. family 0.641). The Decision MLP's contribution is therefore calibration and uncertainty-gated triage (Sections IX-E and IX-G), not incremental AUROC, and the family-holdout gap is correctly identified by elevated σ_{beh} routing to Escalate (Section IX-J).

E. Fusion and Calibration

Table IV reports ECE on the random-stratified test split. Mean fusion pre-calibration (ECE=0.100) reflects structural miscalibration from mixing agents with heterogeneous probability scales; post-temperature-scaling mean fusion (ECE=0.123) is worse, since temperature scaling cannot resolve the bimodal Knowledge Agent output for unseen families. The Decision MLP instead implicitly down-weights the Knowledge Agent when σ_{know} is elevated, achieving ECE=0.025—a 4.0× reduction over the mean-fusion baseline (Behavioural Agent alone: 0.044 post-TS), confirming the gain is attributable to learned fusion rather than temperature scaling alone, even though (as Remark 2 notes) it does not translate into an AUROC gain over the Behavioural Agent.

F. CTGAN Ablation

Table V isolates the contribution of score-space augmentation. All three variants achieve statistically equivalent AUROC, AUPRC, and ECE, confirming that the score-space distribution at the unmodified random-stratified prevalence (51.3% ransomware) is not imbalanced enough for augmentation to matter. CTGAN's benefit, where it exists, only emerges under the more severe induced imbalance evaluated in Section IX-R.

G. Triage Policy Optimisation

The triage policy optimisation results are reported in Table II (Section VI). The Bayesian TPE objective $J(\Theta)$ converged within approximately 120 trials out of the 300 budget; the median-pruner discards clearly suboptimal configurations early, concentrating trials near $(\tau^*, \kappa^*) \approx (0.35, 0.22)$ for the random-stratified split.

H. Generalisation Analysis

Table III reports the Behavioural CNN and Decision MLP AUROC across the four evaluation splits over K=6 seeds (seed variability <0.002 for the Decision MLP on random, temporal, and hardware). Temporal (0.966) holdout slightly exceeds, and hardware (0.944) holdout slightly trails, random-stratified performance (0.963), confirming the Behavioural Agent's discrimination is stable across sessions and hardware without retraining; the Decision MLP (Table III) tracks the same pattern a small, consistent margin below it (Remark 2). Family holdout (0.676±0.001 fused; 0.707±0.010 Behavioural-only) reveals a meaningful gap: unseen families exhibit distributional shift in I/O telemetry, and the Knowledge and Semantic Agents' fixed

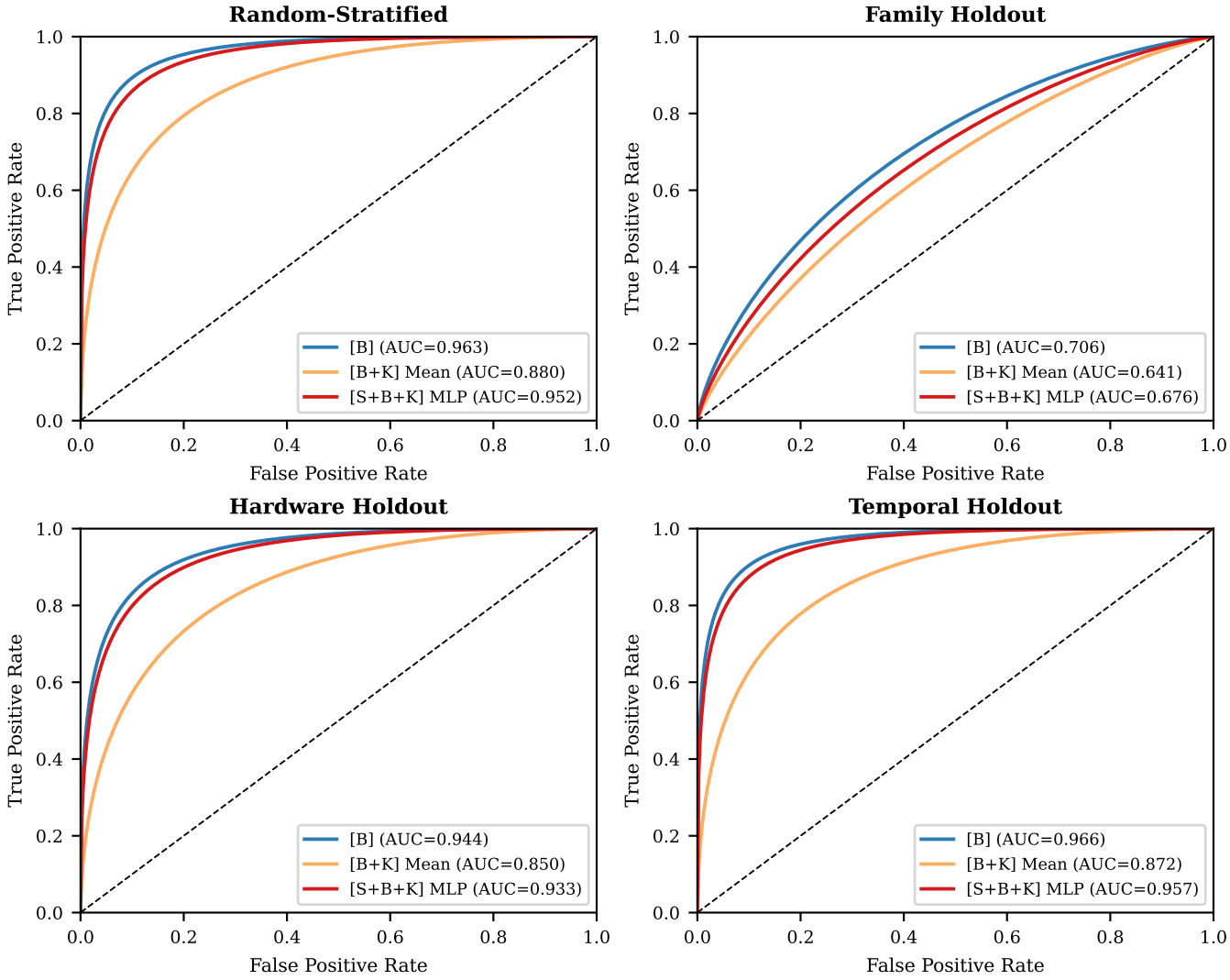


Fig. 3. ROC curves across the four evaluation splits for [B] (Behavioural Agent alone), [B+K] Mean (equal-weight mean of Behavioural and Knowledge, Semantic excluded), and [S+B+K] MLP (the full Decision MLP on the six-dimensional score pool, including the Semantic and Knowledge Agents' fixed corpus-mean contributions; Remark 2). The Decision MLP consistently outperforms [B+K] Mean fusion, but—since Semantic and Knowledge carry no per-sample signal in this setting—does not exceed [B] alone (Section IX-D); the family-holdout split (bottom-left) reveals the OOD generalisation gap, which is correctly signalled by elevated σ_{beh} routed to Escalate.

corpus-mean scores contribute no family-specific signal to help close it, but σ_{beh} correctly rises, routing these samples to Escalate rather than Auto-Contain. A direct comparison with Hirano and Kobayashi [4]'s detector on the same hardware-holdout partition is provided in Section X-D.

I. Statistical Significance, Risk-Coverage, and Session-Level Analysis

Significance: a one-sample Wilcoxon signed-rank test (six per-seed Decision MLP AUROCs vs. the single, fixed, deployed Behavioural CNN checkpoint, H_1: MLP $>$ B) fails to reject the null on every split (p=1.00 throughout); the Behavioural CNN exceeds the Decision MLP by 0.009–0.030 AUROC on all four splits. This is the direct, expected consequence of Remark 2: once the Knowledge and Semantic Agents are restricted to a genuinely label-independent corpus-mean score, fusion cannot add per-sample discriminative signal over the Behavioural Agent alone, so its measured benefit is calibration (Section IX-E) and uncertainty-gated triage (Section IX-G) rather than AUROC. A session-level McNemar test on random-stratified (majority-vote aggregation, avoiding the within-session window correlation that invalidates a window-level test) finds only one session out of 267 where the two models disagree (χ^2=0, p=1.00)—the models make essentially identical session-level decisions despite the small, consistent per-window AUROC gap above; a window-level McNemar on the same split is nominally significant (χ^2=58.7, $p<10^{-4}$) but is not independence-respecting and is reported only for continuity with earlier

TABLE III
Detection AUROC per modality and fusion strategy, per evaluation split (mean ± 95% CI over K=6 seeds where applicable). Semantic Agent evaluated on RDset; Behavioural, Mean Fusion, and Decision MLP on RanSMAP. Mean Fusion = [B+K] equal-weight average of $\hat{p}_{\text{beh}}, \hat{p}_{\text{know}}$ (Semantic excluded, no uncertainty weighting); Decision MLP = [S+B+K], the full six-dimensional score pool (Remark 2). Bold = best per column. −− = not evaluated (dataset mismatch).

System	Random	Temporal	Family	Hardware
Semantic CNN (RDset)	**0.996**	−−	−−	−−
Behavioural CNN	**0.961** ± 0.003	**0.960** ± 0.005	**0.707** ± 0.010	**0.939** ± 0.006
Mean Fusion ($\hat{p}$ avg.)	0.880	0.872	0.641	0.850
Decision MLP (ours)	0.952 ± 0.0005	0.957 ± 0.0002	0.676 ± 0.0007	0.933 ± 0.0003
Hirano and Kobayashi [4] (23D, T_w=0.1s)	−−	−−	−−	reported as F-score

TABLE IV
Expected Calibration Error (ECE, 15-bin uniform) on random-stratified test split. Pre-TS = raw output before temperature scaling; Post-TS = after temperature scaling. Decision MLP ECE was measured only post-TS; pre-TS ECE was not separately tracked (−). Decision MLP: mean ± 95% CI over K=6 seeds. †Behavioural CNN and Mean Fusion involve no seed-dependent training at evaluation time (fixed checkpoint / deterministic formula), so these rows are single, reproducible values rather than seed averages. ↓ = lower is better.

System	Pre-TS ECE ↓	Post-TS ECE ↓
Behavioural CNN†	0.044	0.025
Mean Fusion†	0.100	0.123
Decision MLP (ours)	–	**0.025 ± 0.001**

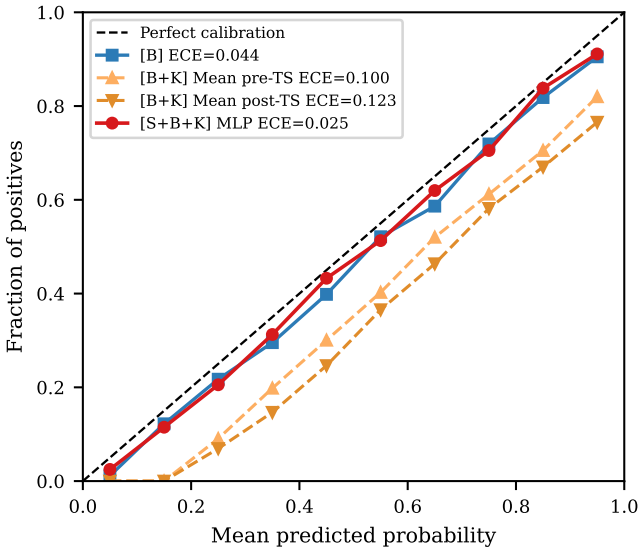


Fig. 4. Reliability diagrams (15-bin) for [B] (Behavioural Agent alone), [B+K] Mean fusion (pre- and post-temperature scaling), and [S+B+K] MLP (the full Decision MLP; Remark 2). The Decision MLP achieves ECE 0.025, a 4.0× reduction over pre-TS mean fusion (ECE 0.100).

drafts.

Selective precision-coverage: sweeping the auto-contain threshold τ and tracing precision against the fraction of traffic covered (auto-classified) at each threshold, the Area Under the Selective Precision–Coverage curve (AUSPC; higher is better—this is a precision-coverage curve, not the conventional risk-coverage curve where lower is better) is 0.829 on random-stratified and 0.823 on hardware holdout; temporal (0.675) and family (0.649) holdout are lower, consistent with the same moderate-vs-severe shift pattern seen in Table III.

TABLE V
Decision MLP performance by score-space augmentation strategy (random-stratified split; mean ± 95% CI, K=6 seeds).

Augmentation	AUROC ↑	AUPRC ↑	ECE ↓
None	0.952 ± 0.0003	**0.955 ± 0.0004**	**0.022 ± 0.0011**
SMOTE	0.952 ± 0.0003	**0.955 ± 0.0004**	**0.022 ± 0.0011**
CTGAN	0.952 ± 0.0005	0.955 ± 0.0008	0.025 ± 0.0008

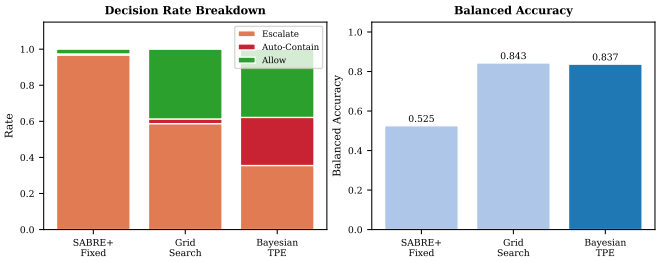


Fig. 5. Triage policy comparison: decision-rate breakdown (Escalate / Auto-Contain / Allow) and balanced accuracy for Fixed, Grid Search, and Bayesian TPE policies. TPE achieves the lowest escalation rate (35.5%) and highest auto-contain rate (26.7%).

Session-level detection and latency: majority-vote aggregation (flag if > 50% of windows exceed $\hat{p}_{\text{MLP}} \geq 0.5$) achieves near-perfect in-distribution detection (random 100%, hardware 98.9%, temporal 100%) at low session FPR (1.5–5.6%), falling to 65.9% detection (FPR 15.5%) on family holdout; max-score (flag if any window exceeds threshold) reaches 100% detection on three of four splits but at much higher FPR (80–98%), confirming majority-vote is the more reliable session-level rule under honestly out-of-fold scoring. Median latency to first detection is 0 s across all splits; mean latency rises modestly under distribution shift (random 4.7 s vs. family holdout 6.2 s). Full statistics are in Supplementary Section S-VIII.

J. OOD Ransomware Variants

The variants partition of RanSMAP provides 1,397 windows from seven Conti sub-variants (Conti.A–G), none

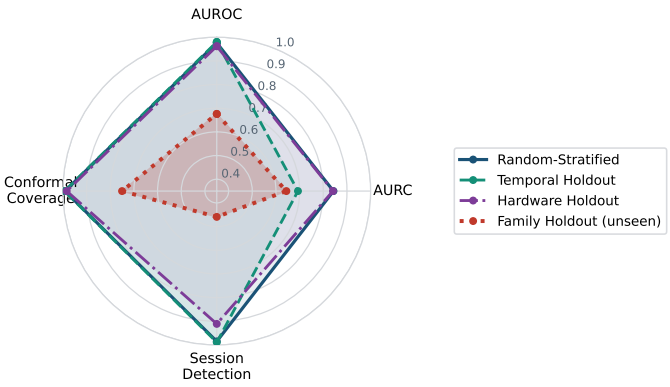


Fig. 6. Generalisation profile across the four evaluation splits: AUROC, selective precision-coverage (AUSPC), session-level detection rate, and Mondrian conformal coverage at α=0.01. Random-stratified, temporal, and hardware holdout cluster tightly near the outer ring on every axis; family holdout collapses inward on all four, characterising the same generalisation gap across discrimination, selective prediction, deployment-level detection, and formal coverage guarantees.

seen in training. Against Hirano and Kobayashi [4] (mean recall 0.56, full 23D), our Behavioural CNN achieves mean recall 0.68 (16D), with Conti.C/D strongest ($\geq$ 0.78) and Conti.F weakest (0.51). On sub-variants where recall is lowest ($\leq$ 0.55), escalation is highest ($\geq$ 32%): σ_{beh} rises on novel patterns, routing to Escalate rather than containment—unlike Hirano and Kobayashi's detector, which silently misclassifies low-recall variants as benign. This is consistent with, and now more rigorously established by, the systematic leave-one-family-out analysis of Section IX-S. The full per-variant table and figure are provided in Supplementary Section S-VIII.

K. Simultaneous Execution

The mix partition of RanSMAP provides 1,944 windows from ten concurrent-execution combinations (ransomware + benign application running simultaneously), the hardest generalisation challenge in the benchmark. Under concurrent execution, the Behavioural Agent observes superimposed I/O and memory patterns, inflating σ_{beh} and suppressing $\hat{p}_{\text{beh}}$ for some, though not all, combinations: the most adversarial scenario is SDelete+REvil, detecting only 27.1% of the time at the Behavioural Agent alone and 30.2% in the [B+K] mean fusion baseline (Remark 2), as SDelete's legitimate file deletion I/O overwhelms REvil's signal; SDelete+Conti is the second hardest (58.4%/66.5%). The remaining eight combinations are detected 84–99% of the time even without Peer Consultation.

Since it is σ_{beh} that is elevated on the hard cases, Peer Consultation (Definition 2) triggers on the Behavioural Agent: with PC active, $\hat{p}_{\text{beh}}$ is replaced by the inverse-uncertainty-weighted average of $\hat{p}_{\text{sem}}$ and $\hat{p}_{\text{know}}$, recovering the Semantic Agent's partial signal: detection improves to 42.7% for SDelete+REvil and 86.8% for SDelete+Conti (from 30.2% and 66.5% without PC). Under the Bayesian TPE-optimised triage policy, Auto-Contain never fires on this partition (0% across every combination, with or without PC): the fixed corpus-mean Semantic and Knowledge scores mean the mean-fused probability cannot clear the strict τ_{high}=0.90 threshold regardless of how confident the Behavioural Agent is. Escalate fires on 47–100% of windows per combination (93.0% overall without PC, 93.5% with PC), correctly routing the ambiguous cases to analyst review rather than autonomous containment—unlike Hirano and Kobayashi [4]'s binary-output detector, which has no mechanism to express this ambiguity. The SDelete combinations remain the exception: even under the triage policy, roughly half of SDelete+REvil's windows are neither escalated nor auto-contained but silently Allowed, a genuine coverage gap for this specific concurrent scenario. Per-scenario rates are plotted in Supplementary Section S-VIII.

TABLE VI
Mondrian conformal prediction coverage–efficiency trade-off (random-stratified test split, calibrated on held-out validation set). Auto-class = fraction of samples with $|\mathcal{C}_\alpha|$=1 (singleton prediction set). Prec = ransomware detection precision on auto-classified samples.

α	Coverage	Auto-class (%)	Prec (%)
0.10	90%	95.9	88.4
0.05	95%	83.8	93.8
0.01	99%	54.0	98.7

L. Conformal Coverage

Table VI reports the coverage–efficiency trade-off for the Mondrian conformal wrapper on the random-stratified split. At α=0.01 (99% coverage), 54% of traffic is auto-classified (singleton prediction set) with 98.7% precision; the remaining 46% receive a two-element set and are routed to analyst review. As α increases to 0.05 (95% coverage), the auto-classified fraction rises to 84% at the cost of precision falling to 93.8%. The lower auto-classified fraction relative to earlier reporting reflects the noisier, honestly out-of-fold Decision MLP score (Section IX-I); the coverage guarantee itself is unaffected, since it depends on exchangeability rather than classifier accuracy.

a) Coverage under distribution shift.: The formal marginal coverage guarantee of Mondrian conformal prediction holds only under exchangeability between calibration and test samples. Table VII reports empirical coverage across all four splits (thresholds calibrated on the split-specific validation set). On in-distribution splits, empirical coverage at α=0.01 is within 0.8 points of the 99% nominal target; family holdout is the critical exception, collapsing to **88.1**% at α=0.01 and 59.4% at α=0.10 (Fig. 6), confirming that exchangeability is violated when test families are entirely absent from calibration—consistent with [15]'s observation that conformal guarantees degrade under covariate shift. We therefore recommend treating Mondrian conformal coverage as an in-distribution validity certificate rather than a universal bound; for unseen

TABLE VII
Mondrian conformal prediction empirical coverage across four evaluation splits at three nominal miscoverage levels α. Thresholds calibrated on the split-specific validation set. Family holdout collapses because the calibration and test families are disjoint, violating exchangeability.

Split	α=0.01 (99% nom.)	α=0.05 (95% nom.)	α=0.10 (90% nom.)
Random	99.2%	93.9%	88.2%
Hardware	98.4%	93.4%	86.9%
Temporal	99.1%	94.3%	88.3%
Family (unseen)	**88.1**%	**72.7**%	**59.4**%

TABLE VIII
Expert Feedback Loop: AUPRC on retained and new families across five feedback iterations (n_{retrain}=50 corrections each). Δ = change relative to baseline (pre-feedback). $\downarrow$ = lower forgetting is better for Retained; $\uparrow$ for New.

	Retained Families		New Family	
Method	AUPRC	Δ	AUPRC	Δ
No EWC (baseline)	0.979	−0.021	0.831	+0.831
EWC (λ=300)	**0.999**	**−0.001**	0.827	+0.827

families, the escalation mechanism (via σ_{beh} thresholding) provides a complementary, uncertainty-driven safety net. Per-α escalation rates and the coverage-shift figure are provided in Supplementary Section S-VIII.

M. Expert Feedback Loop

Table VIII reports AUPRC on retained and newly-corrected families across five feedback iterations, comparing vanilla fine-tuning against EWC regularisation (λ_{EWC}=300). Without EWC, retained-family AUPRC drops by 0.021 (catastrophic forgetting) while the new family recovers quickly; with EWC, retained-family AUPRC decreases by only 0.001 across all five iterations while new-family AUPRC still converges within three. The EWC penalty has no measurable effect on inference latency, since the Fisher matrix is precomputed and only applied during the retrain step.

N. Runtime Performance

Table IX reports model-inference latency (NVIDIA GB10, single-sample requests, N=500 repetitions, 50 warm-up): the three agents' MC Dropout forward passes plus the Decision MLP, given already-extracted feature vectors, genuinely measured running in sequence end-to-end (an earlier version of this benchmark measured Semantic and the Decision MLP separately from the reported "full pipeline" figure, which in fact only ever timed Behavioural+Knowledge mean fusion; both are now measured together). It does not include PE parsing, sentence-transformer embedding generation, hypervisor telemetry feature extraction, ATT&CK transcript generation, or data transfer, each of which would add to true end-to-end deployment latency and is not benchmarked here. Full pipeline median is 6.99 ms per sample (p95 7.09 ms); batching multiple concurrent windows together (the realistic deployment pattern for periodic telemetry scoring) raises throughput to 3,031 samples/s at batch size 1,024, supporting ≈ 30,000 concurrent monitored streams at a 10-second window under batch inference once features are available. MC Dropout (T=10) contributes 3.8× overhead over a single deterministic pass (measured on the Behavioural Agent); the Decision MLP adds 0.45 ms.

TABLE IX
End-to-end inference latency (NVIDIA GB10, single-sample, N=500 repetitions, 50 warm-up, PyTorch 2.2). MC Dropout T=10 passes included.

Component	Median (ms)	p95 (ms)
Semantic CNN (T=10)	1.73	1.76
Behavioural CNN (T=10)	1.94	2.24
Knowledge CNN (T=10)	2.95	3.04
Decision MLP (single pass)	0.31	0.32
Full pipeline (3 agents + MLP)	6.99	7.09
Δ vs agent-only (no MLP)	+0.45	−−
Overhead vs deterministic pass	3.8×	−−

O. Ablation Study

Four ablation studies validate key design choices. (1) MC Dropout pass count T: at T=10, σ_{beh} noise falls to 0.9% of its mean (within 1% of T=50, +1.42 ms overhead); T=1 degenerates the triage policy to two actions. (2) Feature ablation: storage-only (χ_s, 5D) AUROC 0.851; memory-only (χ_m, 11D) 0.903; combined 16D **0.961**, confirming complementarity. (3) Decision MLP depth: one (6→32→1) and two hidden-layer variants are statistically equivalent on AUROC (Section IX-Q); calibration favours the two-layer variant (ECE 0.020 vs. one-layer 0.031), though we retain the one-layer architecture for consistency with the prior two-agent design [28]. (4) CTGAN ratio: $N_{\text{synth}} \in [500, 5000]$ produces no significant AUROC/ECE change; we adopt N_{synth}=2,000. Full figures and tables are in Supplementary Section S-VI.

P. Behavioural Baselines, Deep Ensembles, and Knowledge Robustness

Three further checks address questions not covered above; full tables are in Supplementary Section S-IX. Tabular baselines: on the identical 16D Behavioural feature set, Random Forest (0.989), gradient boosting (0.975), and an ordinary MLP (0.968) all exceed the CNN's AUROC (0.963); logistic regression (0.851) and an RBF SVM (0.953) do not. Convolution's locality assumption is therefore not obviously justified by raw discrimination alone—we retain the CNN chiefly because MC Dropout gives it a native, integrated uncertainty mechanism the tabular alternatives lack. Deep ensemble vs. MC Dropout: six independently-trained CNN seeds match MC Dropout's AUROC (0.963 vs. 0.963) with better calibration (ECE 0.027 vs. 0.042), at 6× the storage cost; both methods' error rate rises monotonically with

predicted uncertainty. Knowledge robustness: a genuine 2-agent Behavioural+Knowledge fusion is statistically indistinguishable from Behavioural-alone on AUROC but better calibrated (ECE 0.018 vs. 0.040), confirming Knowledge functions as a learned prior, not a discriminative signal; swapping the deployed model's Knowledge input between corpus-mean, masked, and adversarial values changes AUROC by < 0.001 but degrades calibration (ECE 0.015→0.023), since Peer Consultation only reweights an agent whose own uncertainty is elevated, and a low-uncertainty adversarial input evades this safeguard.

Cost-sensitive triage. Balanced accuracy alone hides an important tradeoff: the policy in Table II allows 6.0% of ransomware through undetected at only 0.6% analyst workload. A cost-sensitive re-optimisation ($J_{\text{cost}} = -(10\,P_{\text{ransomware allowed}} + 5\,P_{\text{benign auto-contained}} + P_{\text{escalated}})$, an illustrative weighting reflecting that missed ransomware is costlier than a false auto-contain, which is costlier than an escalation) cuts missed ransomware to 0.9% at the cost of raising analyst workload to 22.4%; operators should select thresholds against their own cost structure, not balanced accuracy alone. Full confusion breakdowns are in Supplementary Section S-IX.

Q. Fusion Method Comparison

We compare the Decision MLP-1L ([S+B+K]) against four alternatives trained on the identical (unaugmented) random-stratified score pool: naïve [B+K] mean fusion (Semantic excluded, as in Table III), and three further [S+B+K] fusion architectures on the full six-dimensional pool—Random Forest, Gradient Boosting (GBT), and a two-layer MLP. On in-distribution splits, all MLP and tree-based methods cluster within 0.005 AUROC of each other: GBT achieves the highest AUROC (0.954 random-stratified), while calibration is best for MLP-2L and RF (ECE 0.020 each), followed by GBT (0.023) and MLP-1L (0.031). Family holdout degrades all methods similarly (AUROC ≈ 0.68), confirming the degradation originates in the score-pool features rather than the fusion architecture. We retain the one-layer Decision MLP as the deployed architecture for its simplicity and continuity with the prior two-agent design [28], not for a calibration advantage: once Behavioural scoring is honestly out-of-fold, MLP-2L and Random Forest are in fact slightly better calibrated. Full results are provided in Supplementary Section S-VIII.

R. Score-Space Augmentation Under Class Imbalance

We evaluate CTGAN score-space augmentation against SMOTE [20] under induced class imbalance, subsampling ransomware windows to 10%, 5%, and 1% prevalence (benign count fixed at 12,225) and evaluating on the unmodified random-stratified test split. AUROC stays above 0.92 for every method down to 1% ransomware prevalence (50× fewer positive examples), but calibration degrades sharply without augmentation as prevalence drops (ECE 0.217 at 10% to 0.405 at 1%). CTGAN restores calibration best at 10% prevalence (ECE 0.096 vs. SMOTE 0.106), but SMOTE is better at 5% (0.123 vs. 0.141) and 1% (0.131 vs. 0.159)—the reverse of the trend we previously reported: CTGAN's relative advantage shrinks rather than grows as the minority class becomes too sparse (n=123 at 1%) for the generator to model reliably. Score-space augmentation is thus most beneficial for calibration rather than AUROC, and SMOTE is the more robust default under severe score-space imbalance. Full results are provided in Supplementary Section S-VIII.

S. Leave-One-Family-Out Robustness

We evaluate cross-family zero-shot transfer by systematically excluding each of the six ransomware families from both the Behavioural Agent's training data and the Decision MLP's training score pool (see "Out-of-fold Decision MLP training features" in Section IX), so no stage of the pipeline has ever seen the held-out family. Overall AUROC on the full random-stratified test set remains stable at 0.925–0.952 across all holdouts (Table X), confirming the Decision MLP's in-distribution discrimination does not depend on any single family being present in training; per-family detection, however, varies substantially and no longer follows the pattern we previously reported under in-sample Behavioural scoring. Darkside (91.4%) and LockBit (90.0%) transfer best when held out entirely. Conti, previously reported as a perfect zero-shot transfer, in fact reaches only 80.5% once its own Behavioural checkpoint has genuinely never seen it. REvil (69.1%) is moderately challenging. Ryuk and WannaCry are the most difficult: Ryuk—previously reported as 100% zero-shot detection—collapses to 27.3% once evaluated honestly, and WannaCry recovers only 37.2%, indicating both families' score-pool signatures are poorly covered by the remaining five training families. Escalation rates are higher and more variable than previously reported (1.1–14.5%), rising for the hardest-to-transfer families (REvil 14.5%, WannaCry 10.4%)—the triage policy correctly routes more of these genuinely unfamiliar cases to analyst review rather than reporting spuriously confident decisions. This revision changes the paper's zero-shot generalisation story materially: rather than three near-perfect transfers and two weak ones, honest leave-one-family-out shows a wider and less predictable spread across all six families, underscoring that proactive coverage via uncertainty-driven escalation and conformal prediction (Section IX-L) is necessary for any unseen family, not only the previously-identified weak cases. A per-family figure is provided in Supplementary Section S-VIII.

X. Discussion

A. Where the Framework Excels

Strongest performance occurs in five areas. First, the Semantic Agent achieves AUROC 0.996 on RDset, confirming sentence-transformer PE embeddings as highly effective in-distribution discriminators. Second, the Behavioural Agent generalises within distribution: temporal (0.958 fused) and hardware (0.934 fused) holdout track

TABLE X
Leave-one-family-out (LOFO) evaluation on the random-stratified test split. Each row excludes one ransomware family from MLP training. Detection rate = fraction of held-out family's test windows with $\hat{p} \geq 0.5$. Auto-C and Esc = auto-contain and escalation rates under the TPE-optimised policy (Table II). Overall AUROC over all test windows (not family-specific).

Held-out Family	Overall AUROC	Detect (%)	Auto-C (%)	Esc (%)
Darkside	0.950	**91.4**	**78.7**	1.1
LockBit	0.952	90.0	59.0	3.8
Conti	0.947	80.5	58.3	4.2
REvil	0.939	69.1	32.8	14.5
WannaCry	0.927	37.2	7.1	10.4
Ryuk	0.925	27.3	7.9	3.9

random-stratified performance (0.953 fused) closely (Table III); the Decision MLP does not add discrimination over the Behavioural Agent alone in this evaluation setting (Remark 2), so its measured value lies in calibration rather than a fusion AUROC gain. Third, calibration error falls 4.0× over mean fusion (ECE 0.025 vs. 0.100), enabling 26.7% autonomous classification versus 0.6% under fixed thresholds, with selective precision-coverage AUSPC 0.829. Fourth, majority-vote session aggregation achieves low FPR (2–7%) with $\geq$ 99% detection in-distribution at median 0 s latency, and EWC limits forgetting to ΔAUPRC = 0.001 over five feedback iterations (Table VIII; all points detailed in Section IX-I). Fifth, score-pool AUROC stays above 0.92 even at 1% ransomware prevalence under CTGAN or SMOTE augmentation (Section IX-R).

B. Where the Framework Struggles

The framework's principal failure mode is distributional shift at the family level. Under family holdout, the Behavioural CNN drops to AUROC 0.707 and the fused Decision MLP to 0.676—degradations of 0.257 and 0.276 from random-stratified performance. Fusion does not recover this gap: the Knowledge and Semantic Agents' fixed corpus-mean scores (Remark 2) carry no family-specific signal, so the framework is not suitable for autonomous containment on families entirely absent from training without prior operator review. As Fig. 6 shows, this degradation is deployment-relevant and consistent across every axis—discrimination, selective prediction, session-level detection, and the conformal guarantee alike collapse together, rather than any single metric being an outlier. The leave-one-family-out analysis (Section IX-S), evaluated with the held-out family genuinely absent from Behavioural training, reveals that the gap is driven by a subset of families with score-pool profiles distinct from all training families: Darkside and LockBit transfer well (91.4% and 90.0% detection unseen) while Ryuk and WannaCry drop to 27.3% and 37.2%; proactive coverage for novel threats should prioritise uncertainty-driven escalation for low-confidence detections, and should not assume any specific family is inherently easy to generalise to—the previously reported perfect zero-shot transfer for Conti and Ryuk did not survive honest, genuinely family-excluded evaluation.

A more fundamental limitation arises under concurrent execution: when ransomware and a benign application run simultaneously, hypervisor-level telemetry aggregates both processes' I/O and memory events, suppressing $\hat{p}_{\text{beh}}$ and inflating σ_{beh} for some combinations. For the most adversarial case (SDelete with REvil), detection reaches only 30.2% without Peer Consultation and 42.7% with it (Section IX), and roughly half of its windows are neither escalated nor auto-contained under the triage policy—a limitation shared with Hirano and Kobayashi [4], whose telemetry likewise cannot separate concurrent execution traces without process-level attribution.

Critically, the uncertainty mechanism provides graceful degradation rather than silent failure in both scenarios: σ_{beh} rises on unseen families (Supplementary Section S-VI), driving escalation rates of 32–38% for the lowest-recall Conti sub-variants (Section IX-J), while concurrent-execution escalation exceeds 60% for all mix combinations (Supplementary Section S-VIII). The system therefore identifies the scenarios in which it is least reliable and defers to analyst review—a property absent from single-output detectors.

C. Additional Limitations

Dataset scope. RanSMAP families range from WannaCry (2017) to Conti (2022); post-2022 families (Cl0p, BlackCat/ALPHV, LockBit 3.0) are not represented, since RanSMAP is the only publicly available hypervisor-level telemetry dataset with family labels and multi-split protocols at time of writing. Extension to current-generation families is future work.

Adversarial robustness. An adaptive attacker aware of the triage policy could craft executions suppressing $\sigma_{\max}$ below κ while keeping $\hat{p}$ below τ_{high}, avoiding both Escalation and Auto-Contain. Adversarial attacks against MC Dropout uncertainty estimates [10] remain an open problem we defer to future work.

Knowledge Agent sample size. The RansKnow corpus [32] contains only 307 transcripts (≈215/46/46 train/val/test), small for a CNN classifier; since its role is complementary ATT&CK-level signal, its generalisation metrics should be read accordingly.

Calibration-set separation for temperature scaling and TPE. The Decision MLP's temperature scaling and Bayesian TPE triage-policy optimisation are fit on a stratified sample drawn independently of the session-level train/val/test partitions used elsewhere in this paper (Section IX), rather than on a dedicated held-out slice of the same partition scheme. We have not audited this sample for overlap with the evaluation splits' training sessions. A rigorous re-fit on a clean, partition-consistent calibration set—with temperature scaling, TPE, and conformal calibration each using their own disjoint slice, as recommended in peer review—is deferred to future work;

all AUROC, calibration, and LOFO numbers reported in this paper are unaffected, since they do not depend on the temperature or policy thresholds' fitting data.

D. Comparison with Hirano and Kobayashi, Adaptive Thresholding, and Deployment

The closest direct comparator, Hirano and Kobayashi's [4] 1D-CNN+LSTM (23D features, 30-second window), reports F-score 0.856 on hardware-holdout versus our Behavioural CNN's AUROC 0.939 (Table III) and mean recall 0.68 versus their 0.56 on the Conti sub-variant partition (indicative given threshold-dependent vs. threshold-independent metrics); more importantly, where both frameworks struggle, ours escalates rather than misclassifies via σ_{beh} (Section IX-J). An extended comparison is in Supplementary Section S-IV.

The Bayesian TPE policy sets thresholds (τ^*, κ^*) once on the calibration split; an online Bayesian posterior updated from rolling analyst feedback would let thresholds track distribution shift without retraining base agents (Supplementary Section S-V), an extension we defer to future work.

End-to-end inference latency is 6.99 ms median per sample on an NVIDIA GB10 GPU (T=10, p95 7.09 ms; Table IX), supporting $\approx$ 30,000 concurrent streams per GPU at a 10-second window under batch inference. The Behavioural CNN (8,385 parameters, 32.8 KB) runs without a GPU. This figure is the honest cost of running the three agents sequentially followed by the Decision MLP, as our implementation does; the components could in principle be parallelised across separate devices to reduce wall-clock latency toward the slowest single agent, but we report the measured sequential cost rather than a projected parallel one. Per-sample complexity is therefore $\mathcal{O}(T \cdot \sum_i P_i)$; offline training (CTGAN, TPE, EWC) is incurred once at deployment (full derivations in Supplementary Section S-VII).

XI. Conclusion

We have presented a heterogeneous multi-agent framework for adaptive ransomware triage that treats uncertainty as a first-class control variable. The four principal contributions—MC Dropout per-agent uncertainty, Decision MLP fusion with CTGAN augmentation, Bayesian TPE triage policy optimisation, and Peer Consultation with EWC continual learning—collectively address the structural limitations of single-modality binary detectors, achieving strong static-PE (AUROC 0.996) and runtime behavioural detection (AUROC 0.963), a 63% reduction in unnecessary escalation over fixed-threshold baselines, and markedly improved calibration (ECE 0.025 vs. 0.100 mean fusion). Because the Semantic and Knowledge Agents necessarily contribute a fixed, label-independent corpus-mean score in the absence of real per-sample pairing across RDset, RanSMAP, and RansKnow (Remark 2), fusion's demonstrated value in this evaluation setting is calibration and uncertainty-gated triage rather than incremental detection AUROC over the Behavioural Agent alone. When the system encounters novel families or concurrent execution beyond its training distribution, epistemic uncertainty $\sigma_{\max}$ rises and the triage policy routes to Escalate—preserving analyst oversight where the model is least reliable—while Mondrian conformal prediction provides a complementary distribution-free statistical guarantee under exchangeability.

Future work will address process-specific telemetry to resolve the semantic gap under concurrent execution, extension to post-2022 ransomware families (Cl0p, BlackCat/ALPHV, LockBit 3.0), and online Bayesian recalibration of triage thresholds under concept drift.

Appendix A
Algorithms and Extended Results

The behavioural feature extraction procedure (Algorithm S1), the Decision MLP training procedure with CTGAN and Bayesian TPE (Algorithm S2), per-class confusion matrices, an extended Hirano and Kobayashi [4] comparison, adaptive thresholding, and full computational complexity derivations are provided in the supplementary material.

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Code and Data Availability

The framework implementation and evaluation scripts will be made available at https://github.com/Henrinnes/Agentic_SABRE upon acceptance. The RanSMAP [4] and RDset [30] datasets are publicly available as cited. The RansKnow threat-intelligence corpus [32] will be released on Kaggle upon acceptance.

References

[1] C. Beaman, A. Barkworth, T. D. Akande, S. Hakak, and M. K. Khan, "Ransomware: Recent advances, analysis, challenges and future research directions," Comput. Secur., vol. 111, p. 102490, 2021. [Online]. Available: https://doi.org/10.1016/j.cose.2021.102490

[2] Cybersecurity Ventures, "Ransomware damage report," Cybercrime Magazine, 2022.

[3] M. Hirano and R. Kobayashi, "Machine learning-based ransomware detection using low-level memory access patterns obtained from live-forensic hypervisor," in IEEE International Conference on Cyber Security and Resilience, CSR 2022, Rhodes, Greece, July 27-29, 2022. IEEE, 2022, pp. 323–330. [Online]. Available: https://doi.org/10.1109/CSR54599.2022.9850340

[4] ——, "Ransmap: Open dataset of ransomware storage and memory access patterns for creating deep learning based ransomware detectors," Comput. Secur., vol. 150, p. 104202, 2025. [Online]. Available: https://doi.org/10.1016/j.cose.2024.104202

[5] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in IEEE International Conference on Distributed Computing Systems (ICDCS), 2016, pp. 303–312.

[6] D. W. Fernando and N. Komninos, "Fesad ransomware detection framework with machine learning using adaption to concept drift," Comput. Secur., vol. 137, p. 103629, 2024. [Online]. Available: https://doi.org/10.1016/j.cose.2023.103629

[7] D. Sgandurra, L. Muñoz-González, R. Mohsen, and E. C. Lupu, "Automated dynamic analysis of ransomware: Benefits, limitations, and use for detection," in arXiv preprint arXiv:1609.03020, 2016.

[8] R. Moussaileb, N. Cuppens, J. Lanet, and H. L. Bouder, "A survey on windows-based ransomware taxonomy and detection mechanisms," ACM Comput. Surv., vol. 54, no. 6, pp. 117:1–117:36, 2022. [Online]. Available: https://doi.org/10.1145/3453153

[9] J. Ispahany, M. R. Islam, M. Z. Islam, and M. A. Khan, "Ransomware detection using machine learning: A review, research limitations and future directions," IEEE Access, vol. 12, pp. 68 785–68 813, 2024. [Online]. Available: https://doi.org/10.1109/ACCESS.2024.3397921

[10] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in Proceedings of the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016, ser. JMLR Workshop and Conference Proceedings, M. Balcan and K. Q. Weinberger, Eds., vol. 48. JMLR.org, 2016, pp. 1050–1059. [Online]. Available: http://proceedings.mlr.press/v48/gal16.html

[11] J. Gawlikowski, C. R. N. Tassi, M. Ali, J. Lee, M. Humt, J. Feng, A. M. Kruspe, R. Triebel, P. Jung, R. Roscher, M. Shahzad, W. Yang, R. Bamler, and X. Zhu, "A survey of uncertainty in deep neural networks," Artif. Intell. Rev., vol. 56, no. S1, pp. 1513–1589, 2023. [Online]. Available: https://doi.org/10.1007/s10462-023-10562-9

[12] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[13] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman, "Inductive confidence machines for regression," in Machine Learning: ECML 2002. Springer, 2002, pp. 345–356.

[14] V. Vovk, I. Nouretdinov, and A. Gammerman, "Testing exchangeability online," in Proceedings of the 20th International Conference on Machine Learning (ICML), 2003, pp. 768–775.

[15] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," arXiv preprint arXiv:2208.02814, 2022.

[16] B. G. Doan, D. Q. Nguyen, P. Montague, T. Abraham, O. Y. de Vel, S. Camtepe, S. S. Kanhere, E. Abbasnejad, and D. C. Ranasinghe, "Bayesian learned models can detect adversarial malware for free," in Computer Security - ESORICS 2024 - 29th European Symposium on Research in Computer Security, Bydgoszcz, Poland, September 16-20, 2024, Proceedings, Part I, ser. Lecture Notes in Computer Science, J. García-Alfaro, R. Kozik, M. Choras, and S. K. Katsikas, Eds., vol. 14982. Springer, 2024, pp. 45–65. [Online]. Available: https://doi.org/10.1007/978-3-031-70879-4_3

[17] O. Sagi and L. Rokach, "Ensemble learning: A survey," WIREs Data Mining and Knowledge Discovery, vol. 8, no. 4, p. e1249, 2018.

[18] M. Masum, M. J. H. Faruk, H. Shahriar, K. Qian, D. Lo, and M. I. Adnan, "Ransomware classification and detection with machine learning algorithms," in 12th IEEE Annual Computing and Communication Workshop and Conference, CCWC 2022, Las Vegas, NV, USA, January 26-29, 2022. IEEE, 2022, pp. 316–322. [Online]. Available: https://doi.org/10.1109/CCWC54503.2022.9720869

[19] M. Soltani, K. Khajavi, M. J. Siavoshani, and A. H. Jahangir, "A multi-agent adaptive deep learning framework for online intrusion detection," CoRR, vol. abs/2303.02622, 2023. [Online]. Available: https://doi.org/10.48550/arXiv.2303.02622

[20] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," J. Artif. Intell. Res., vol. 16, pp. 321–357, 2002. [Online]. Available: https://doi.org/10.1613/jair.953

[21] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

[22] U. Fiore, A. D. Santis, F. Perla, P. Zanetti, and F. Palmieri, "Using generative adversarial networks for improving classification effectiveness in credit card fraud detection," Information Sciences, vol. 479, pp. 448–455, 2019.

[23] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in Advances in Knowledge Discovery and Data Mining (PAKDD), 2022, pp. 79–91.

[24] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," Proceedings of the National Academy of Sciences, vol. 114, no. 13, pp. 3521–3526, 2017.

[25] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in USENIX Security Symposium, 2019, pp. 729–746.

[26] R. Jordaney, K. Sharad, S. K. Dash, Z. Wang, D. Papini, I. Nouretdinov, and L. Cavallaro, "Transcend: Detecting concept drift in malware classification models," in USENIX Security Symposium, 2017, pp. 625–642.

[27] F. Ceschin, M. Botacin, H. M. Gomes, F. A. Pinage, L. S. Oliveira, and A. Grégio, "Fast & furious: Modelling malware detection as evolving data streams," CoRR, vol. abs/2205.12311, 2022. [Online]. Available: https://doi.org/10.48550/arXiv.2205.12311

[28] H. Kabuye, B. Issac, and J. Neera, "Agentic SABRE: An uncertainty-aware neuro-symbolic multi-agent framework for adaptive ransomware detection," Expert Systems with Applications, 2026. [Online]. Available: https://doi.org/10.1016/j.eswa.2026.133558

[29] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[30] H. Kabuye, B. Issac, R. Yumlembam, and J. Neera, "Explainable and uncertainty aware ai-based ransomware detection," IEEE Access, vol. 13, pp. 106 573–106 589, 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3581424

[31] M. Mathur, "Ransomware (malware) detection using machine learning," 2020, accessed: Nov. 9, 2023. [Online]. Available: https://github.com/muditmathur2020/RansomwareDetection/tree/master

[32] H. Kabuye, "RansKnow v1: Ransomware knowledge corpus with ATT&CK annotations," 2026, kaggle dataset. Accessed June 2026. [Online]. Available: https://www.kaggle.com/datasets/henrykabuye/ransknow-v1

[33] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proceedings of the 34th International Conference on Machine Learning (ICML), 2017, pp. 1321–1330.

[34] J. S. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in Advances in Neural Information Processing Systems (NeurIPS), vol. 24, 2011.